```
15          pBrush->SetColor(color);
16          D2D1_ELLIPSE ellipse = D2D1::Ellipse(D2D1::Point2F(cx, cy),
                radius, radius);
17          pRT->DrawEllipse(ellipse, pBrush, 1.0f);
18          return;
19      }
20
21      // 计算外圆和内圆半径
22      float outerRadius = radius + width / 2.0f;
23      float innerRadius = radius - width / 2.0f;
24      if (innerRadius < 0) innerRadius = 0;
25
26      // 获取Direct2D工厂
27      ID2D1Factory* pFactory = nullptr;
28      pRT->GetFactory(&pFactory);
29
30      // 创建路径几何对象
31      ID2D1PathGeometry* pPathGeometry = nullptr;
32      pFactory->CreatePathGeometry(&pPathGeometry);
33
34      // 打开几何接收器
35      ID2D1GeometrySink* pSink = nullptr;
36      pPathGeometry->Open(&pSink);
37
38      // 绘制外圆（顺时针）
39      pSink->BeginFigure(D2D1::Point2F(cx + outerRadius, cy),
40                         D2D1_FIGURE_BEGIN_FILLED);
41      D2D1_ARC_SEGMENT arc;
42      arc.size = D2D1::SizeF(outerRadius, outerRadius);
43      arc.rotationAngle = 0.0f;
44      arc.sweepDirection = D2D1_SWEEP_DIRECTION_CLOCKWISE;
45      arc.arcSize = D2D1_ARC_SIZE_SMALL;
46      arc.point = D2D1::Point2F(cx - outerRadius, cy);
47      pSink->AddArc(arc);
48      arc.point = D2D1::Point2F(cx + outerRadius, cy);
49      pSink->AddArc(arc);
50      pSink->EndFigure(D2D1_FIGURE_END_CLOSED);
51
52      // 绘制内圆（逆时针，形成环形）
53      if (innerRadius > 0) {
```